

A GRADED CONSTRUCTION OF THE FREE δ_p -RING OVER THE SPHERE

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ABSTRACT. We study spherical lifts of free δ_p -rings, as asked in Question 3 of Carmeli–Nikolaus–Yuan. We describe a weightwise construction of a p -complete Moore E_∞ -ring with degree-zero homotopy the completed free δ_p -ring on one generator. The construction separates the prescribed Frobenius operations from the obstruction theory of graded commutative ring spectra. Its homotopical ingredient is a duality calculation for the Mitchell–Priddy summands,

$$[\Sigma^j L(k), \mathbb{S}_p^\wedge] \cong \pi_j L(k) \oplus \pi_j L(k-1), \quad L(-1) = 0.$$

We explain the cell attachments, their compatibility with power operations, and the passage from the p -complete to an integral Moore lift.

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1. INTRODUCTION

Let p be a prime. A Moore E_∞ -ring is a commutative ring spectrum R for which $R \otimes H\mathbb{Z}$ is concentrated in degree zero. Carmeli, Nikolaus and Yuan construct a natural $\widehat{\delta}$ -ring structure on $\pi_0 R$ by means of the Tate-valued Frobenius [CNY26, Proposition 2.41]. Their Question 3 asks whether the free δ_p -ring on one generator has a spherical lift. The question is not merely whether one can put a multiplication on a Moore spectrum: the multiplication must be coherently E_∞ , and its induced Frobenius operation must be the prescribed one.

We use the conventions

$$\mathbb{Z}\{x\} = \mathbb{Z}[x_0, x_1, \dots], \quad \delta_p(x_i) = x_{i+1}, \quad \varphi(x_i) = x_i^p + px_{i+1}.$$

Write $A_p = (\mathbb{Z}_p[x_0, x_1, \dots])_p^\wedge$, with the continuous extension of this δ_p -structure. A statement about the completion of a polynomial algebra always refers to the p -adic completion of the entire algebra, not to an unrestricted formal power series ring in the x_i .

Theorem 1.1 (Main result). *For each prime p there is a p -complete Moore E_∞ -ring R_p and an isomorphism of δ_p -rings*

$$\pi_0 R_p \cong A_p,$$

where the operation on $\pi_0 R_p$ is that of [CNY26, Proposition 2.41]. There is also an integral Moore E_∞ -ring $R_{\mathbb{Z}}$ with $\pi_0 R_{\mathbb{Z}} \cong \mathbb{Z}\{x\}$ such that its p -completed δ_p -structure is the given free structure.

The local construction is weightwise: there is a connected graded commutative algebra $\mathcal{R}$ in p -complete spectra, with $\mathcal{R}_0 \simeq \mathbb{S}_p^\wedge$, each $\mathcal{R}_w$ a finite wedge of copies of $\mathbb{S}_p^\wedge$, and classes $x_i \in \pi_0 \mathcal{R}_{p^i}$ satisfying

$$H\mathbb{F}_p \otimes \mathcal{R} \simeq H(\mathbb{F}_p[x_0, x_1, \dots]), \quad \delta_p(x_i) = x_{i+1}.$$

Where the obstruction occurs. The weight convention assigns weight p^i to x_i . In a graded E_∞ -algebra B , the products of components of weights $< w$ determine a latching object $L_w(B)$, and a choice of the component B_w amounts to a map $L_w(B) \rightarrow B_w$. The cofiber is the weight- w indecomposable spectrum $\mathrm{TAQ}_w(B)$.

For the spherical monoid ring $\mathbb{S}[\mathbb{N}]$, these indecomposables are governed by the stable symmetric-power filtration. After p -completion they are supported in weights $1, p, p^2, \dots$. The weight-1 term is $\mathbb{S}_p^\wedge$, while the weight- p^k term is $\Sigma^k L(k)$, where $L(k)$ is the Mitchell–Priddy spectrum and $L(0) = \mathbb{S}_p^\wedge$; see Theorem 2.5. In particular, the first indecomposable at weight 1 is the sphere itself. Retaining the suspension exponent k is essential for the later excess-filtration calculation.

At weight p^n , the lower-stage algebra has a free summand, accounting for the desired monomials, and an excess spectrum filtered by suspensions $\Sigma^{k-1} L(k)$, $1 \leq k \leq n$. The new class x_n is introduced by replacing this excess with a free sphere summand. The obstruction to making the necessary projections compatible belongs to

$$[\Sigma^{k-2} L(k), \mathbb{S}_p^\wedge], \quad k \geq 2. \quad (1)$$

The cohomotopy calculation in Section 5 shows that these groups vanish. This is the only stable-homotopy vanishing used in the induction; the $k = 1$ term is treated by choosing an attaching map rather than setting it equal to zero.

How the Frobenius is imposed. For an element $a \in \pi_0 R$ in a p -complete Moore E_∞ -ring, the total Σ_p -power operation has the form

$$P_{\Sigma_p}(a) = a^p \varepsilon_{\Sigma_p} + \delta_p(a)(p \varepsilon_{\Sigma_p} - \mathrm{tr}_{\Sigma_p}). \quad (2)$$

Let V be the reduced summand of $(\Sigma_+^\infty B \Sigma_p)_p^\wedge$, and put $t_{\Sigma_p} = (\mathrm{tr}_{\Sigma_p} - p \varepsilon_{\Sigma_p})|_V$. Since t_{Σ_p} is not torsion, restriction of (2) to V detects $\delta_p(a)$. At the new weight the attaching map is chosen to restrict to $-t_{\Sigma_p}$, and this forces $\delta_p(x_{n-1}) = x_n$; see Proposition 3.4 and Theorem 4.7.

The proofs are organized around these two separate tasks. Section 2 gives the weightwise algebra and the symmetric-power input. Section 3 treats the Frobenius and transfer operations. Section 4 constructs the graded algebra by successive attachments. Section 5 proves (1), and Section 6 passes to completed and integral spherical lifts.

Conventions. Write Sp for spectra, $\mathrm{Sp}_p^\wedge$ for p -complete spectra, and $\mathbb{S}_p^\wedge$ for the p -complete sphere. Components of graded objects are written with subscripts; square brackets $X[w]$ place X in weight w , while $\Sigma^j X$ denotes suspension. All tensor products of p -complete spectra are derived and p -completed. We put $[X, Y] = \pi_0 \mathrm{Map}(X, Y)$ and use $\delta_p(a) = (\varphi(a) - a^p)/p$ when the coefficient ring is p -torsion free. A finite free $\mathbb{S}_p^\wedge$ -module means a finite wedge of *unsuspended* copies of $\mathbb{S}_p^\wedge$.

2. GRADED ALGEBRAS AND WEIGHTWISE INDECOMPOSABLES

2.1. Truncation and extension in one weight. Let $\mathcal{C}$ be a stable presentably symmetric monoidal ∞ -category whose tensor product preserves colimits separately. We give

$$\mathrm{Gr}(\mathcal{C}) = \mathrm{Fun}(\mathbb{N}^{\mathrm{disc}}, \mathcal{C})$$

the Day-convolution tensor product; hence $(X \otimes Y)_w \simeq \bigoplus_{a+b=w} X_a \otimes Y_b$. Write $\mathrm{Gr}_{\leq n}(\mathcal{C})$ for objects supported in weights $0, \dots, n$. A commutative algebra B is *connected in weight* if B_0 is the tensor unit $\mathbf{1}$.

Lemma 2.1. *Weight truncation $\tau_{\leq n}$ is a symmetric monoidal localization, with the truncated tensor product given by $\tau_{\leq n}(X \otimes Y)$. The same holds between two consecutive weight bounds.*

Proof. A morphism is inverted by $\tau_{\leq n}$ precisely when it is an equivalence in weights $\leq n$. In each such weight, $X \otimes Z$ only uses summands X_a with $a \leq n$, so tensoring preserves the localized equivalences. The criterion for a monoidal localization in [Lur17, §2.2.1] applies. Since a map from X into an object supported in weights $\leq n$ depends only on $\tau_{\leq n}X$, the asserted adjunction has the stated orientation. $\square$

Let $i_n : \mathrm{CAlg}(\mathrm{Gr}_{\leq n}(\mathcal{C})) \rightarrow \mathrm{CAlg}(\mathrm{Gr}_{\leq n+1}(\mathcal{C}))$ denote the left adjoint to truncation (we suppress the truncation functor from the notation). For an algebra A supported in weights $\leq n$, set

$$L_{n+1}(A) = (i_n A)_{n+1}.$$

For a larger algebra B , the counit produces $\varepsilon_{B, n+1} : L_{n+1}(\tau_{\leq n} B) \rightarrow B_{n+1}$. Thus L_{n+1} contains all weight- $(n+1)$ operations already prescribed by the lower weights.

Proposition 2.2 (Extension in one weight). *Let A be connected in weights $\leq n$, and let $f : L_{n+1}(A) \rightarrow M$ be a map in $\mathcal{C}$. There exists a connected algebra B in weights $\leq n+1$ whose truncation is A , whose new component is M , and whose latching map is f . Its space of maps to an algebra B' is the homotopy pullback*

$$\mathrm{Map}(B, B') \simeq \mathrm{Map}(A, \tau_{\leq n} B') \times_{\mathrm{Map}(L_{n+1}(A), B'_{n+1})} \mathrm{Map}(M, B'_{n+1}),$$

where the right-hand map is precomposition with f . Every extension of A occurs in this way.

Proof. Write $L = L_{n+1}(A)$ and let $\mathrm{Free}_{\leq n+1}(Z[n+1])$ be the free algebra on Z concentrated in the new weight. Form

$$B = i_n A \sqcup_{\mathrm{Free}_{\leq n+1}(L[n+1])} \mathrm{Free}_{\leq n+1}(M[n+1]).$$

A free generator of weight $n+1$ contributes only in weights 0 and $n+1$ after truncation. In the two-sided bar model for this pushout, every term involving two positive-weight generators has weight $> n+1$. Consequently, the weights $\leq n$ are those of A , and the new weight is the ordinary pushout of $L \xleftarrow{\mathrm{id}} L \xrightarrow{f} M$, namely M . The induced map from the latching object is f .

To identify maps from B , map the displayed pushout into B' . Adjunction identifies maps from its first vertex with maps $A \rightarrow \tau_{\leq n} B'$ and maps from each free vertex with maps from its generator into B'_{n+1} . This gives the stated homotopy pullback, including its compatibility condition. Applying the construction to the latching map of any given extension produces an equivalence in every weight, so there are no other extensions. $\square$

2.2. Weightwise TAQ.

Definition 2.3. For a connected graded algebra B and $w \geq 1$, set

$$\mathrm{TAQ}_w(B) = \mathrm{cofib}(L_w(\tau_{<w}B) \xrightarrow{\varepsilon_{B,w}} B_w).$$

The direct sum of these spectra in their respective weights is denoted $\mathrm{TAQ}(B)$.

The notation reflects the following square-zero universal property. In this statement $\mathbf{1} \oplus M[w]$ is the augmented algebra in which the positive-weight ideal has square zero.

Proposition 2.4. *For a connected graded algebra B the following hold.*

(i) *For any $M \in \mathcal{C}$,*

$$\mathrm{Map}_{\mathrm{CAlg}}(B, \mathbf{1} \oplus M[w]) \simeq \mathrm{Map}_{\mathcal{C}}(\mathrm{TAQ}_w(B), M).$$

In particular TAQ_w preserves colimits of connected algebras.

- (ii) *If X is concentrated in positive weights, then $\mathrm{TAQ}_w(\mathrm{Free}(X)) \simeq X_w$.*
- (iii) *$\mathrm{TAQ}_w(B)$ depends only on $\tau_{\leq w}B$. It commutes with the symmetric monoidal base changes used here, including reduction modulo p and derived p -completion in the bounded-below cases at hand.*
- (iv) *If $B \rightarrow B'$ induces an equivalence on TAQ_w for $w \leq n$, then it is an equivalence in weights $\leq n$.*
- (v) *Rescaling all weights by d gives*

$$\mathrm{TAQ}_v(r_d B) \simeq \begin{cases} \mathrm{TAQ}_{v/d}(B), & d \mid v, \\ 0, & d \nmid v. \end{cases}$$

Proof. A map from B to $\mathbf{1} \oplus M[w]$ has a unique restriction in weights $< w$, namely the augmentation. By Proposition 2.2, choosing the remaining part of such a map means choosing $B_w \rightarrow M$ together with a null-homotopy of its composite with $L_w(\tau_{<w}B)$. In a stable category this is precisely a map from the cofiber $\mathrm{TAQ}_w(B)$ to M , proving (i). Since this identification exhibits TAQ_w as a left adjoint, it preserves colimits.

For a free algebra, maps to a square-zero target are maps from the generating object; (i) then gives (ii) by Yoneda. Both the latching map and its cofiber use only weights at most w , giving the first part of (iii). The formulas for free algebras, followed by colimit preservation, prove the base-change assertion on a simplicial resolution by free algebras; the bounded-below hypotheses ensure that the indicated completion is derived completion and has no unaccounted inverse-limit term.

For (iv), argue by induction on w . In weight 0 both algebras are $\mathbf{1}$. If their lower truncations agree, so do their latching objects. Comparing the two cofiber sequences

$$L_w \longrightarrow B_w \longrightarrow \mathrm{TAQ}_w(B), \quad L_w \longrightarrow B'_w \longrightarrow \mathrm{TAQ}_w(B')$$

shows that $B_w \rightarrow B'_w$ is an equivalence. Finally, r_d is symmetric monoidal, sends a generator of weight u to one of weight du , and preserves square-zero extensions. Apply (i) to obtain (v). $\square$

2.3. The spherical monoid ring and the power-of- p weights. Give $\mathbb{S}[\mathbb{N}] = \Sigma_+^\infty \mathbb{N}$ the grading in which the monoid generator t has weight 1. We shall use the symmetric-power description of its derived indecomposables. Write $\mathrm{SP}^m(S^r)$ for the ordinary m -th symmetric product of the pointed space S^r . By contrast, $\mathrm{SP}^m(\mathbb{S})$ will always mean the *spectrum-level* symmetric power in the filtration of $H\mathbb{Z}$, obtained by stabilizing these space-level symmetric

products. In particular, $\mathrm{SP}^m(\mathbb{S})$ is not the suspension spectrum of $\mathrm{SP}^m(S^0)$. In the usual Mitchell–Priddy normalization, set

$$L(k) = \Sigma^{-k}(\mathrm{SP}^{p^k}(\mathbb{S})/\mathrm{SP}^{p^k-1}(\mathbb{S}))_p^\wedge, \quad k \geq 1, \quad L(0) = \mathbb{S}_p^\wedge.$$

Theorem 2.5 (Symmetric-power calculation). *There is a natural equivalence*

$$\mathrm{TAQ}_w(\mathbb{S}[\mathbb{N}]) \simeq \mathrm{colim}_r \Sigma^{-r} \Sigma^\infty(\mathrm{SP}^w(S^r)/\mathrm{SP}^{w-1}(S^r)).$$

After p -completion it has the following form:

$$\mathrm{TAQ}_w(\mathbb{S}[\mathbb{N}])_p^\wedge \simeq \begin{cases} \mathbb{S}_p^\wedge, & w = 1, \\ \Sigma^k L(k), & w = p^k, \quad k \geq 1, \\ 0, & w > 1 \text{ is not a power of } p. \end{cases} \quad (3)$$

Proof. The first statement is the stabilization of the iterated reduced bar resolution of the augmented commutative monoid algebra: its r -fold indecomposable layer in weight w is the quotient of the w -th symmetric product of S^r by the $(w-1)$ -st. The suspension maps in the bar resolution are the standard stabilization maps in the symmetric-power filtration. This is the symmetric-power model used in the Whitehead filtration; compare [Nak58, AD01, MP83, AL10].

For $w = 1$ the quotient is S^r , so the colimit is $\mathbb{S}$; this case should not be included in the suspended $L(k)$ formula. For $w > 1$, Nakaoka’s mod- p calculation identifies the nonzero layers precisely at powers p^k . The corresponding stabilized layer is $\Sigma^k L(k)$ in the normalization above; this is the symmetric-power layer comparison of [AL10, §2.1] and [MP83]. Note that the exponent k is part of the identification, not merely a shift in notation. Bounded-below p -complete spectra with zero mod- p homology vanish, giving the third case. $\square$

Lemma 2.6. *For $k \geq 1$ put $c(k) = 2p^k - 2 - k$. Then*

$$\pi_j L(k) = 0 \quad (j < c(k)).$$

In particular $c(k) \geq k$ and $c(k-1) > k-2$ for $k \geq 2$.

Proof. The first possible nonzero mod- p homology degree in the p^k symmetric-power layer is $2p^k - 2$. In the classical symmetric-power filtration this follows from the admissible-word calculation; see [AL10, §2.1 and the proof of Proposition 4.11] and [Kuh15, §1]. Desuspension by k gives the bound $c(k)$. Since $L(k)$ is bounded below and p -complete, its first nonzero homotopy group, if any, would map nontrivially to mod- p homology by the lowest-degree Hurewicz argument. Thus homotopy has the same lower bound. The inequalities reduce to $2p^k - 2 \geq 2k$ and $2p^{k-1} - 2 - (k-1) > k-2$; the latter is immediate for $k = 2$ and increases with k . $\square$

Corollary 2.7 (No unforced weights). *Suppose B is a connected graded E_∞ -algebra in p -complete spectra, every B_w is a finite free $\mathbb{S}_p^\wedge$ -module, and*

$$\pi_0(H\mathbb{F}_p \otimes B) \cong \mathbb{F}_p[x_0, x_1, \dots], \quad |x_i| = p^i.$$

Then $\mathrm{TAQ}_w(B) = 0$ for non- p -power weights w . Consequently $L_w(\tau_{<w} B) \rightarrow B_w$ is an equivalence in these weights.

Proof. Because each component is a wedge of unsuspended spheres, $H\mathbb{F}_p \otimes B$ is degreewise discrete. Its graded E_∞ -algebra structure is consequently the Eilenberg–Mac Lane structure attached to the displayed graded polynomial ring. It is the tensor product, over $H\mathbb{F}_p$, of copies of $H\mathbb{F}_p \otimes \mathbb{S}[\mathbb{N}]$ with weights rescaled by p^i . By Proposition 2.4 and Theorem 2.5, its

indecomposables are supported at weights $p^i p^k$. Reduction of $\mathrm{TAQ}_w(B)$ modulo p therefore vanishes outside powers of p . The relevant spectra are bounded below and derived p -complete, so the indecomposable itself vanishes. $\square$

3. FROBENIUS OPERATIONS IN A GRADED CONSTRUCTION

The distinction between a graded algebra and a graded δ_p -ring is worth making explicit. A Frobenius lift sends a homogeneous element of weight w to one of weight pw ; its associated p -derivation has the same property on homogeneous elements, although δ_p is not additive on arbitrary sums.

3.1. The Tate-valued Frobenius. Let $\mathcal{R} = \bigoplus_{w \geq 0} \mathcal{R}_w$ be a connected graded algebra with each component finite free over $\mathbb{S}_p^\wedge$. Put $U = \bigoplus_w \mathcal{R}_w$ after forgetting the grading and $R_U = U_p^\wedge$.

Proposition 3.1. *The ring R_U is a p -complete Moore E_∞ -ring, and*

$$\pi_0 R_U \cong \left(\bigoplus_{w \geq 0} \pi_0 \mathcal{R}_w \right)_p^\wedge.$$

The Tate–Frobenius induces operations on $\pi_0 U$ satisfying

$$\varphi(\pi_0 \mathcal{R}_w) \subseteq \pi_0 \mathcal{R}_{pw}, \quad \delta_p(\pi_0 \mathcal{R}_w) \subseteq \pi_0 \mathcal{R}_{pw}.$$

The δ_p -structure of R_U is the p -adically continuous extension of this operation.

Proof. As a spectrum, U is a wedge of p -complete spheres. It is connective and $H\mathbb{Z} \otimes U$ is discrete; the same remains true after p -completion by [CNY26, Lemma 2.40]. This proves the Moore condition. The formula for $\pi_0 R_U$ follows from the same lemma and the fact that the degree-zero homotopy of U is a direct sum of finite free $\mathbb{Z}_p$ -modules.

For a fixed homogeneous input in $\mathcal{R}_w$, the Tate diagonal

$$\mathcal{R}_w \longrightarrow (\mathcal{R}_w^{\otimes p})^{tC_p}$$

followed by graded multiplication lands in $(\mathcal{R}_{pw})^{tC_p}$. The trivial-action comparison $\mathcal{R}_{pw} \simeq (\mathcal{R}_{pw})^{tC_p}$ from [CNY26, Proposition 2.36], applicable to a finite free sphere module, therefore gives a map $\varphi_w : \mathcal{R}_w \rightarrow \mathcal{R}_{pw}$. On π_0 these maps agree with the Frobenius lift of the underlying ring. Since a^p also has weight pw , the identity $p\delta_p(a) = \varphi(a) - a^p$ and p -torsion-freeness of $\pi_0 \mathcal{R}_{pw}$ imply the second inclusion. All constructions are natural for $U \rightarrow U_p^\wedge$, and the corresponding map on degree-zero homotopy is p -adic completion. The induced operation is therefore its continuous extension. $\square$

Lemma 3.2 (Recognizing the free structure). *Let*

$$B = \mathbb{Z}_p[y_0, y_1, \dots], \quad |y_i| = p^i,$$

carry a p -torsion-free graded δ_p -structure, and put $I = B_{>0}$. For each i there is a unique $c_i \in \mathbb{Z}_p$ such that

$$\delta_p(y_i) \equiv c_i y_{i+1} \pmod{I^2}.$$

The unique δ_p -map $\mathbb{Z}_p\{x\} \rightarrow B$ sending x_0 to y_0 is an isomorphism if and only if all c_i are units. Equivalently, B is the free graded δ_p -ring on its weight-one generator precisely when all c_i are units.

Proof. The only indecomposable monomial in weight p^{i+1} is y_{i+1} , so $B_{p^{i+1}} = \mathbb{Z}_p y_{i+1} \oplus (I^2)_{p^{i+1}}$. The identities for a p -derivation show that $\delta_p(I^2) \subseteq I^2$: for $a, b \in I$,

$$\delta_p(ab) = a^p \delta_p(b) + b^p \delta_p(a) + p \delta_p(a) \delta_p(b),$$

and each term belongs to I^2 , since $\delta_p(I) \subseteq I$ by homogeneity. The p -derivation of a sum changes modulo I^2 by no cross term in positive-weight elements. Moreover, φ is the identity on the coefficient ring $\mathbb{Z}_p$, whence

$$\delta_p(uy_i) \equiv u \delta_p(y_i) \pmod{I^2}, \quad u \in \mathbb{Z}_p.$$

Iterating gives $\delta_p^i(y_0) \equiv (c_0 \cdots c_{i-1}) y_i \pmod{I^2}$. If all c_i are units, then the substitution $x_i \mapsto \delta_p^i(y_0)$ is triangular with invertible diagonal in every finite weight and is thus an isomorphism weight by weight. Conversely, an isomorphism induces an isomorphism of the indecomposable modules I/I^2 , so none of these coefficients can be a nonunit. $\square$

3.2. The transfer class. For a finite group G write $\varepsilon_G \in [\Sigma_+^\infty BG, \mathbb{S}_p^\wedge]$ for the class of the constant map. A finite G -set T also determines a transfer class, denoted t_T . We use $\text{tr} = t_{C_p}$ for the translation C_p -set and $\text{tr}_{\Sigma_p} = t_{\{1, \dots, p\}}$ for the tautological Σ_p -set.

Proposition 3.3. *There are additive descriptions*

$$[\Sigma_+^\infty BC_p, \mathbb{S}_p^\wedge] = \mathbb{Z}_p \varepsilon \oplus \mathbb{Z}_p \text{tr}, \quad \text{tr}^2 = p \text{tr},$$

and

$$[\Sigma_+^\infty B\Sigma_p, \mathbb{S}_p^\wedge] = \mathbb{Z}_p \varepsilon_{\Sigma_p} \oplus \mathbb{Z}_p \text{tr}_{\Sigma_p}.$$

If $(\Sigma_+^\infty B\Sigma_p)_p^\wedge \simeq \mathbb{S}_p^\wedge \vee V$ is its basepoint splitting, then

$$[V, \mathbb{S}_p^\wedge] \cong \mathbb{Z}_p t_{\Sigma_p}, \quad t_{\Sigma_p} = (\text{tr}_{\Sigma_p} - p \varepsilon_{\Sigma_p})|_V.$$

In particular t_{Σ_p} has infinite additive order.

Proof. For C_p the degree-zero case of the Segal conjecture identifies stable cohomotopy with the completed Burnside ring. The latter has basis the trivial orbit and the free orbit, and the product of two free orbits is the disjoint union of p free orbits. This gives the first displayed formula.

Let $H = C_p$ be a Sylow subgroup of $G = \Sigma_p$. The suspension transfer for the finite covering $BH \rightarrow BG$ yields a map $\text{trf} : \Sigma_+^\infty BG \rightarrow \Sigma_+^\infty BH$. Its composite with the map induced by the covering is a mod- p homology equivalence: on ordinary mod- p homology the composite is multiplication by the degree $[G : H] = (p-1)!$. Both suspension spectra are connective. After p -completion the composite is therefore an equivalence, by the mod- p Whitehead criterion for connective p -complete spectra. In particular, the completed suspension spectrum of BG is a retract of that of BH , and restriction

$$[\Sigma_+^\infty BG, \mathbb{S}_p^\wedge] \longrightarrow [\Sigma_+^\infty BH, \mathbb{S}_p^\wedge]$$

is injective. (This argument uses the ordinary homology transfer; it does not claim that the corresponding stable composite is literally the degree map.)

The restrictions of ε_{Σ_p} and tr_{Σ_p} are ε and tr , respectively, because the tautological p -point Σ_p -set restricts to a free transitive C_p -set. They therefore map to a basis of the target. Injectivity shows that they form a basis of the source. The basepoint splitting separates ε_{Σ_p} , while $\text{tr}_{\Sigma_p} - p \varepsilon_{\Sigma_p}$ lies in the reduced summand. Its restriction has a nonzero free-orbit coefficient, so the resulting generator t_{Σ_p} is not torsion. $\square$

3.3. The total power operation. Let $P_{\Sigma_p}(a)$ denote the total degree-zero Σ_p -power operation of an element a , viewed as a stable map out of $\Sigma_+^\infty B\Sigma_p$. Write $P_{C_p}(a)$ for its restriction to C_p .

Proposition 3.4 (Transfer formula). *Suppose R is a connective p -complete Moore E_∞ -ring with p -torsion-free $\pi_0 R$, and the power operation is evaluated in a finite free $\mathbb{S}_p^\wedge$ -module summand (as happens for the homogeneous components used below). Then*

$$\begin{aligned} P_{C_p}(a) &= \varphi(a)\varepsilon - \delta_p(a) \operatorname{tr} = a^p \varepsilon + \delta_p(a)(p\varepsilon - \operatorname{tr}), \\ P_{\Sigma_p}(a) &= a^p \varepsilon_{\Sigma_p} + \delta_p(a)(p\varepsilon_{\Sigma_p} - \operatorname{tr}_{\Sigma_p}). \end{aligned}$$

On the reduced summand V this says

$$P_{\Sigma_p}(a)|_V = -\delta_p(a)t_{\Sigma_p}. \quad (4)$$

For $a \in \pi_0 \mathcal{R}_w$ the equality takes place in weight pw .

Proof. By [NS18, Corollary IV.1.10 and Lemma IV.1.20], passage from homotopy fixed points to Tate fixed points sends the total C_p -power operation to the Tate-valued Frobenius. Using the trivial-action equivalence of [CNY26, Proposition 2.36], this says that the image of $P_{C_p}(a)$ under the canonical map to Tate fixed points is represented by $\varphi(a)\varepsilon$.

The norm cofiber sequence shows that the difference is in the image of transfer. In degree zero for a connective coefficient spectrum, π_0 of the homotopy-orbit term is generated over $\pi_0 R$ by the orbit of the unit; thus the difference is $b \operatorname{tr}$ for some $b \in \pi_0 R$. The non-equivariant restriction of $P_{C_p}(a)$ is a^p , whereas the non-equivariant restrictions of ε and tr are 1 and p . Therefore $a^p = \varphi(a) + pb$, and p -torsion-freeness gives $b = -\delta_p(a)$. By Proposition 3.3, restriction from Σ_p to C_p is injective, so the Σ_p formula follows. Restricting to V eliminates ε_{Σ_p} and yields (4). Finally, the construction is homogeneous by Proposition 3.1. $\square$

Proposition 3.5 (First nontrivial weight of the monoid algebra). *Let $h : \operatorname{Free}(\mathbb{S}_p^\wedge[1]) \rightarrow \mathbb{S}_p^\wedge[\mathbb{N}]$ classify the monoid generator t . It is an equivalence below weight p . In weight p it is the projection*

$$(\Sigma_+^\infty B\Sigma_p)_p^\wedge \simeq \mathbb{S}_p^\wedge \vee V \longrightarrow \mathbb{S}_p^\wedge.$$

Hence $\operatorname{TAQ}_p(\mathbb{S}_p^\wedge[\mathbb{N}]) \simeq \Sigma V$ and $\delta_p(t) = 0$.

Proof. For $w < p$ the order of Σ_w is prime to p , so $(\Sigma_+^\infty B\Sigma_w)_p^\wedge \simeq \mathbb{S}_p^\wedge$. In weight p , the map is induced by the collapse $B\Sigma_p \rightarrow *$, hence is the basepoint projection. Its cofiber is ΣV . The monoid Frobenius takes t to t^p ; equivalently, the total power operation has zero restriction to V . Formula (4) and the non-torsion of t_{Σ_p} imply $\delta_p(t) = 0$. $\square$

4. CELL ATTACHMENTS AND THE INDUCTIVE CONSTRUCTION

We now work throughout in connected graded algebras over $\mathbb{S}_p^\wedge$. The point of retaining the grading is that a cell attached in weight w cannot alter any component of smaller weight. This allows the homotopy-theoretic obstruction to be isolated before the choice of the next δ_p -generator.

4.1. Two facts about graded pushouts.

Lemma 4.1 (Filtration of graded modules). *Let K be a connected augmented graded algebra with augmentation $\varepsilon : K \rightarrow \mathbb{S}_p^\wedge$. Every graded K -module M has a filtration by weights, and*

the layer in weight r is the K -module obtained from $M_r[r]$ by restriction along ε . In particular, for a right K -module P and a spectrum Z ,

$$P \otimes_K \varepsilon^*(Z[r]) \simeq (P \otimes_K \mathbb{S}_p^\wedge) \otimes Z[r].$$

If P is supported in nonnegative weights and M in weights $\geq r$, then

$$(P \otimes_K M)_v \simeq 0 \quad (v < r), \quad (P \otimes_K M)_r \simeq P_0 \otimes M_r.$$

Proof. Filter M by its truncations $\tau_{\geq r} M$. On the quotient $\tau_{\geq r} M / \tau_{\geq r+1} M$, an element of K of positive weight acts trivially, because its action would land in a different weight. The K -action therefore factors through $K_0 \simeq \mathbb{S}_p^\wedge$, which identifies the layer with $\varepsilon^*(M_r[r])$. The tensor-product identity follows by associativity of relative tensor products. The first potentially nonzero term in the two-sided bar construction for $P \otimes_K M$ has total weight r , and its only contribution in that weight comes from $P_0 \otimes M_r$. This gives the last two assertions. $\square$

Lemma 4.2 (Attachment in a single weight). *Let E be connected, and let $f : M \rightarrow E_w$ and $g : M \rightarrow N$ be maps of p -complete spectra. Write*

$$E' = E \bigotimes_{\text{Free}(M[w])} \text{Free}(N[w]).$$

Then $E \rightarrow E'$ is an equivalence below weight w ; the square

$$\begin{array}{ccc} M & \xrightarrow{f} & E_w \\ g \downarrow & & \downarrow \\ N & \longrightarrow & E'_w \end{array}$$

is cocartesian in spectra; and

$$\text{TAQ}(E') \simeq \text{TAQ}(E) \sqcup_{M[w]} N[w],$$

where $M[w] \rightarrow \text{TAQ}(E)$ is induced by f .

Proof. The two-sided bar construction for the relative tensor product has no occurrence of a positive-weight generator in total weight $< w$. At weight w , exactly one such generator can occur; consequently its bar construction reduces to the displayed pushout. For the indecomposables, apply colimit preservation and the free-algebra calculation of Proposition 2.4. Both claims are statements about actual homotopy pushouts, so the null-homotopies needed for subsequent attachments are part of the pushout data. $\square$

Lemma 4.3 (Filling a non- p -power weight). *Let B be defined below weight N , with all components finite free, and assume that its mod- p reduction agrees there with the graded polynomial algebra on generators of p -power weights. If N is not a power of p , then $L_N(B)$ is finite free. It reduces modulo p to the vector space spanned by the weight- N monomials, in degree zero. Taking the latching map in weight N to be the identity is the unique extension having $\text{TAQ}_N = 0$.*

Proof. Let P be the corresponding graded polynomial Eilenberg–Mac Lane algebra. The latching object only depends on weights $< N$, so

$$H\mathbb{F}_p \otimes L_N(B) \simeq L_N(\tau_{< N} P).$$

By Corollary 2.7 and the defining cofiber sequence, the right side is equivalent to P_N , a degree-zero $\mathbb{F}_p$ -vector space with its monomial basis. The latching object is connective and

derived p -complete: it is built from bounded-below finite free lower-weight components by finite colimits and homotopy-orbit constructions at fixed weight. Lifting a basis of $\pi_0(P_N)$ produces a map from a finite wedge of $\mathbb{S}_p^\wedge$'s to $L_N(B)$ which becomes an equivalence modulo p . The p -complete Nakayama lemma makes it an equivalence. The assertion about the extension is Proposition 2.2; when the latching map is an equivalence there is no additional choice of indecomposables. $\square$

4.2. The basic blocks. Let $i_{<m,!}$ denote left Kan extension (left adjoint to weight truncation) from algebras in weights $< m$ to all nonnegative weights. Define

$$E^{(j)} = i_{<p^{j+1},!}(\tau_{<p^{j+1}}\mathbb{S}_p^\wedge[\mathbb{N}]), \quad E^{(j)} \xrightarrow{c_j} \mathbb{S}_p^\wedge[\mathbb{N}].$$

The useful feature of $E^{(j)}$ is that it has all the relations of the monoid algebra in the initial range, but no artificially added indecomposable cells above that range.

Proposition 4.4 (Block calculation). *For $j \geq 0$ the following hold.*

- (i) c_j is an equivalence in weights $< p^{j+1}$, and $\mathrm{TAQ}_w(E^{(j)}) = 0$ in weights $w \geq p^{j+1}$.
- (ii) $E^{(0)} \simeq \mathrm{Free}(\mathbb{S}_p^\wedge[1])$, and $E_p^{(0)} \simeq \mathbb{S}_p^\wedge \vee V$.
- (iii) There is a natural splitting

$$E_{p^{j+1}}^{(j)} \simeq \mathbb{S}_p^\wedge \vee \Lambda^{(j)}, \quad \Lambda^{(j)} \simeq \Sigma^j L(j+1),$$

in which $(c_j)_{p^{j+1}}$ projects to the sphere summand.

- (iv) If one kills $\Lambda^{(j)}$ in weight p^{j+1} , then

$$E^{(j)} \bigotimes_{\mathrm{Free}(\Lambda^{(j)}[p^{j+1}])} \mathbb{S}_p^\wedge \simeq E^{(j+1)}. \quad (5)$$

The same identities hold after rescaling all weights by any p -power.

Proof. The counit of the adjunction defining $i_{<m,!}$ is an equivalence below m . The square-zero universal property of Proposition 2.4 shows that a left extension from weights $< m$ has no indecomposables in weights $\geq m$: a map to a square-zero object in such a weight is already determined by the lower-weight map and hence is trivial. This proves (i).

For $j = 0$, Proposition 3.5 gives the claim directly. At $w = p^{j+1}$, the map $c_j : E^{(j)} \rightarrow \mathbb{S}_p^\wedge[\mathbb{N}]$ is an equivalence below w . Comparing their two latching cofiber sequences, and using $\mathrm{TAQ}_w(E^{(j)}) = 0$, gives

$$\mathrm{cofib}(E_w^{(j)} \rightarrow \mathbb{S}_p^\wedge) \simeq \mathrm{TAQ}_w(\mathbb{S}_p^\wedge[\mathbb{N}]) \simeq \Sigma^{j+1} L(j+1).$$

Thus the fiber of the displayed map is $\Sigma^j L(j+1)$. The multiplication of the weight-one generator gives a map $\mathbb{S}_p^\wedge \rightarrow E_w^{(j)}$, and its composite with $E_w^{(j)} \rightarrow \mathbb{S}_p^\wedge$ is the identity: both represent the monomial t^w . This *actual section*, rather than an appeal merely to connectivity, proves the splitting in (iii).

Put $E' = E^{(j)} \bigotimes_{\mathrm{Free}(\Lambda^{(j)}[w])} \mathbb{S}_p^\wedge$. The new weight- w component is $\mathbb{S}_p^\wedge$, and the lower weights are unchanged by Lemma 4.2. In weight w , the indecomposable cofiber becomes $\Sigma \Lambda^{(j)} \simeq \Sigma^{j+1} L(j+1)$, agreeing with Theorem 2.5. In all weights strictly between w and p^{j+2} the indecomposables vanish, as they did for $E^{(j)}$; above this range both E' and $E^{(j+1)}$ have no new indecomposables by their free-extension descriptions. The induced comparison $E' \rightarrow E^{(j+1)}$ is therefore an equivalence on each indecomposable spectrum and hence on all weights by Proposition 2.4(iv). This proves (5). Rescaling is covered by Proposition 2.4(v). $\square$

For $n \geq 1$ and $0 \leq i \leq n-1$ abbreviate

$$E_i^{[n]} = r_{p^i} E^{(n-1-i)}, \quad \Lambda_i^{[n]} = \Sigma^{n-1-i} L(n-i).$$

Thus $E_i^{[n]}$ is generated in weight p^i , agrees with the rescaled monoid algebra below weight p^n , and

$$(E_i^{[n]})_{p^n} \simeq \mathbb{S}_p^\wedge \vee \Lambda_i^{[n]}. \quad (6)$$

For $i = n-1$, the excess summand is $\Lambda_{n-1}^{[n]} \simeq L(1) \simeq V$.

4.3. The projection problem. The stable-homotopy input will be stated separately from the graded construction.

Definition 4.5. The prime p satisfies condition (V_p) if

$$[\Sigma^{k-2} L(k), \mathbb{S}_p^\wedge] = 0 \quad (k \geq 2). \quad (V_p)$$

We will establish (V_p) for every prime in Corollary 5.5. The following observation explains how such a negative suspension group enters an extension of maps; it is convenient to record the direction of the connecting morphism.

Lemma 4.6 (Lifting a projection). *Suppose $X \rightarrow B \xrightarrow{q} C$ is a cofiber sequence in $\mathrm{Sp}_p^\wedge$, and $X \simeq \mathbb{S}_p^{\wedge \vee a} \vee \Lambda$. Suppose also that C has a filtration whose subquotients are finite wedges of $\mathbb{S}_p^\wedge$ and spectra $\Sigma^{k-1} L(k)$ with $k \geq 2$. Under (V_p) , the projection $X \rightarrow \mathbb{S}_p^{\wedge \vee a}$ extends to a map $s : B \rightarrow \mathbb{S}_p^{\wedge \vee a}$. The map*

$$B \xrightarrow{(q,s)} C \oplus \mathbb{S}_p^{\wedge \vee a}$$

is compatible with the given quotient map q on its first coordinate. If C has a projection onto a free sphere summand inducing an isomorphism on π_0 , the resulting B has such a projection as well.

Proof. Applying $[-, \mathbb{S}_p^{\wedge \vee a}]$ to the cofiber sequence gives the exact segment

$$[B, \mathbb{S}_p^{\wedge \vee a}] \longrightarrow [X, \mathbb{S}_p^{\wedge \vee a}] \xrightarrow{\partial} [\Sigma^{-1} C, \mathbb{S}_p^{\wedge \vee a}].$$

The last group is zero. It vanishes for a free sphere layer because $\pi_{-1} \mathbb{S}_p^\wedge = 0$, and for a layer $\Sigma^{k-1} L(k)$ it is the group in (V_p) . Exact sequences for the finite filtration give the vanishing for C . Thus the projection of X extends to s . Taking (q, s) is an explicit way to retain compatibility with the old quotient; no independent homotopy between two subsequently chosen projections is required. The map on π_0 is the direct sum of the given projection on C and the projection of the free summand of X , hence is an isomorphism after choosing the monomial bases. This proves the last assertion. $\square$

4.4. Inductive data and the new weight. For each $n \geq 1$ consider a diagram

$$W(n, -1) = \mathbb{S}_p^\wedge, \quad W(n, 0), \dots, W(n, n-1),$$

with maps $E_i^{[n]} \rightarrow W(n, i)$ and equivalences

$$W(n, i) \otimes_{E_i^{[n]}} \mathbb{S}_p^\wedge \simeq W(n, i-1), \quad 0 \leq i \leq n-1. \quad (7)$$

The image of the generator of $E_i^{[n]}$ is denoted x_i . Write $R^{(n-1)} = W(n, n-1)$. We require that these be cell algebras supported by cells of weight at most p^{n-1} and, below weight p^n , that

$$H\mathbb{F}_p \otimes W(n, i) \simeq H(\mathbb{F}_p[x_0, \dots, x_i]_{< p^n}) \quad \text{in weights } < p^n, \quad (8)$$

with finite free components, and that $\delta_p(x_i) = x_{i+1}$ in $R^{(n-1)}$ whenever $i < n - 1$. Formula (8) for the intermediate quotients is included to make explicit the lower-weight information used in the projection calculation.

The initial stage is

$$W(1, 0) = \text{Free}(\mathbb{S}_p^\wedge[x_0; 1]), \quad W(1, -1) = \mathbb{S}_p^\wedge.$$

Its quotient by the weight-one generator is $\mathbb{S}_p^\wedge$; below p it is the polynomial algebra on x_0 , so it satisfies the stated properties.

Theorem 4.7 (Inductive attachment). *Assume (V_p) . Given stage- n data, one can construct stage- $(n + 1)$ data. More precisely, there is a connective excess spectrum K_n in weight p^n , a map $h : K_n \rightarrow \mathbb{S}_p^\wedge$, and an attachment*

$$R^{(n)} = R^{(n-1)} \otimes_{\text{Free}(K_n[p^n])} \text{Free}(\mathbb{S}_p^\wedge[x_n; p^n]) \quad (9)$$

where $K_n \rightarrow \mathbb{S}_p^\wedge[x_n; p^n]$ is h times the new generator, such that

- (i) $R^{(n-1)} \rightarrow R^{(n)}$ is an equivalence below weight p^n ;
- (ii) each component of $R^{(n)}$ below p^{n+1} is finite free;
- (iii) in weights $< p^{n+1}$, reduction modulo p is $H(\mathbb{F}_p[x_0, \dots, x_n])$;
- (iv) $\delta_p(x_i) = x_{i+1}$ for $0 \leq i < n$;
- (v) the new quotient algebras have the form (7) with n replaced by $n + 1$.

Proof. We give the construction in four parts. The filtration calculation is stated first because it determines both which map may be chosen and why its choice imposes the correct Frobenius relation.

1. *The excess filtration.* For $0 \leq i \leq n - 1$, form the homotopy fiber in weight p^n of the quotient map in (7):

$$X_i \longrightarrow W(n, i)_{p^n} \xrightarrow{q_i} W(n, i - 1)_{p^n}.$$

Let $J_i = \text{fib}(E_i^{[n]} \rightarrow \mathbb{S}_p^\wedge)$, regarded as a module over $E_i^{[n]}$. The relative tensor-product fiber sequence identifies

$$X_i \simeq (W(n, i) \otimes_{E_i^{[n]}} J_i)_{p^n}.$$

Filter J_i from above by its weight truncations $\tau_{\geq v} J_i$. The module-filtration identity of Lemma 4.1 gives the associated graded pieces

$$\text{gr}_v X_i \simeq W(n, i - 1)_{p^n - v} \otimes (J_i)_v, \quad 1 \leq v \leq p^n. \quad (10)$$

Here $(J_i)_v \simeq (E_i^{[n]})_v$ for $v > 0$. If $v < p^n$, the second factor is a finite free sphere module, because the block agrees with the monoid algebra below its first excess weight. The first factor is finite free by the inductive condition (8). Thus all the layers with $v < p^n$ are finite free spheres. The final layer, $v = p^n$, uses $W(n, i - 1)_0 \simeq \mathbb{S}_p^\wedge$, so (6) makes it

$$\text{gr}_{p^n} X_i \simeq \mathbb{S}_p^\wedge \vee \Lambda_i^{[n]}.$$

This filtration actually splits as a filtration of spectra, though we do not use a canonical splitting. At each step we have a cofiber sequence $F^{v+1} X_i \rightarrow F^v X_i \rightarrow \text{gr}_v X_i$ with a free-sphere quotient. Starting with the connective top layer, all $F^{v+1} X_i$ are connective; therefore an extension is classified by a map from a free sphere to $\Sigma F^{v+1} X_i$, which vanishes because $\pi_{-1} F^{v+1} X_i = 0$. Splitting successively yields

$$X_i \simeq \mathbb{S}_p^{\wedge \vee a_i} \vee \Lambda_i^{[n]}. \quad (11)$$

In (10), a summand from $(J_i)_v$ corresponds to a positive power of x_i of weight v , while the other factor records a monomial in $x_0, \dots, x_{i-1}$. Thus a_i is exactly the number of weight- p^n monomials involving x_i . The inclusion of the top filtration layer singles out $\Lambda_i^{[n]}$, and because the filtration came from the relative tensor product, its maps are compatible with the successive quotient maps in (7).

2. Compatible free projections. Start with $W(n, -1)_{p^n} = 0$. Suppose we have already constructed

$$\rho_{i-1} : W(n, i-1)_{p^n} \longrightarrow \mathbb{S}_p^{\wedge \vee N_{i-1}}$$

which is an isomorphism on π_0 and whose fiber $K(n, i-1)$ is filtered by the layers $\Lambda_0^{[n]}, \dots, \Lambda_{i-1}^{[n]}$. The layers in this range are $\Sigma^{k-1}L(k)$ with $k = n, n-1, \dots, n-i+1 \geq 2$; hence condition (V_p) and connectivity of $\mathbb{S}_p^\wedge$ imply

$$[\Sigma^{-1}W(n, i-1)_{p^n}, \mathbb{S}_p^\wedge] = 0.$$

Apply Lemma 4.6 to (11). We obtain a map $\sigma_i : W(n, i)_{p^n} \rightarrow \mathbb{S}_p^{\wedge \vee a_i}$ extending the projection of X_i onto its free part. Define *the entire projection*, rather than just its homotopy class, by

$$\rho_i = (\rho_{i-1} \circ q_i, \sigma_i) : W(n, i)_{p^n} \longrightarrow \mathbb{S}_p^{\wedge \vee (N_{i-1} + a_i)}.$$

This definition makes the first coordinate agree with the preceding projection by construction. Its map on π_0 is an isomorphism in the monomial bases; put $N_i = N_{i-1} + a_i$ and $K(n, i) = \text{fib}(\rho_i)$. Applying the 3×3 lemma to the morphism between the defining fiber sequences of ρ_i and ρ_{i-1} gives a cofiber sequence

$$\Lambda_i^{[n]} \longrightarrow K(n, i) \longrightarrow K(n, i-1), \quad K(n, -1) = 0. \quad (12)$$

The chosen first-coordinate maps also provide the maps of these sequences needed in the next stage. In particular, the construction does not rely on uniqueness of null-homotopies: the quotient maps and the projections have been chosen together.

3. The attaching map and the δ_p -relation. Take $K_n = K(n, n-1)$. Since the last layer in (12) is $\Lambda_{n-1}^{[n]} \simeq V$, we have a cofiber sequence

$$V \longrightarrow K_n \longrightarrow K(n, n-2).$$

For $n = 1$ the last term is 0. For $n > 1$ it is filtered by $\Sigma^{k-1}L(k)$ with $k \geq 2$. Applying $[-, \mathbb{S}_p^\wedge]$ to the sequence shows that

$$[K_n, \mathbb{S}_p^\wedge] \longrightarrow [V, \mathbb{S}_p^\wedge]$$

is surjective: its connecting homomorphism takes values in $[\Sigma^{-1}K(n, n-2), \mathbb{S}_p^\wedge] = 0$ by (V_p) . Choose a lift

$$h : K_n \longrightarrow \mathbb{S}_p^\wedge, \quad h|_V = -t_{\Sigma_p}.$$

The inclusion $K_n \rightarrow R_{p^n}^{(n-1)}$ determines the left-hand map in (9), and h determines the right-hand map. By Lemma 4.2, weights below p^n do not change. Moreover, ρ_{n-1} has a section: choose lifts of the monomial basis of $\pi_0 \mathbb{S}_p^{\wedge \vee N_{n-1}}$ in $\pi_0 R_{p^n}^{(n-1)}$; the resulting composite with ρ_{n-1} is an invertible matrix over $\mathbb{Z}_p$ and hence a sphere-module equivalence. We may therefore write

$$R_{p^n}^{(n-1)} \simeq \mathbb{S}_p^{\wedge \vee N_{n-1}} \vee K_n.$$

The pushout of $K_n \rightarrow R_{p^n}^{(n-1)}$ along $h : K_n \rightarrow \mathbb{S}_p^\wedge[x_n]_{p^n}$ replaces the second summand by the new sphere. Hence

$$R_{p^n}^{(n)} \simeq \mathbb{S}_p^{\wedge \vee(N_{n-1}+1)}.$$

Modulo p , the new basis vector is x_n and the others are precisely the old monomials of weight p^n . Apply Lemma 4.3 successively for $p^n < w < p^{n+1}$. Each latching object is a free sphere module on the expected monomials. This constructs the algebra through the claimed range, with polynomial mod- p reduction and all components finite free.

It remains to verify that the new degree-zero class has the *required* p -derivation, rather than merely an unspecified one. The last block $E_{n-1}^{[n]}$ is free on x_{n-1} in weight p^{n-1} . In weight p^n , the reduced part of its Σ_p -power operation is represented by V , and the new attachment sends it to $-x_n t_{\Sigma_p}$. By (4) it is also equal to $-\delta_p(x_{n-1})t_{\Sigma_p}$. The two coefficients are elements of the free degree-zero weight- p^n component; Proposition 3.3 shows that multiplication by t_{Σ_p} there is injective. Therefore

$$\delta_p(x_{n-1}) = x_n.$$

For $i < n - 1$, the identities are inherited from the preceding stage by naturality of the Tate–Frobenius.

4. *Propagation of the quotient diagram.* For $0 \leq i < n$ define

$$W(n+1, i) = W(n, i) \otimes_{\text{Free}(K(n, i)[p^n])} \mathbb{S}_p^\wedge, \quad W(n+1, n) = R^{(n)}. \quad (13)$$

The map $E_i^{[n]} \rightarrow W(n, i)$ carries its excess summand $\Lambda_i^{[n]}$ into $K(n, i)$ by (12). Thus, after the quotient in (13), this summand is null. By Proposition 4.4(iv), the map from $E_i^{[n]}$ factors through the next block $E_i^{[n+1]}$.

We check the quotient identities, rather than assuming that the factorization suffices. Since the projections ρ_i were chosen with first coordinate $\rho_{i-1}q_i$, their fibers fit into the *specified* cofiber sequence (12). First impose the relation killing $\Lambda_i^{[n]}$ in the block, and then impose the relation killing all of $K(n, i)$. Associativity of homotopy pushouts and the cofiber sequence $\Lambda_i^{[n]} \rightarrow K(n, i) \rightarrow K(n, i-1)$ identify the remaining attachment, after quotienting by $E_i^{[n+1]}$, with the attachment killing $K(n, i-1)$ in $W(n, i-1)$. In formulas this identifies the two iterated homotopy pushouts as

$$W(n+1, i) \otimes_{E_i^{[n+1]}} \mathbb{S}_p^\wedge \simeq W(n, i-1) \otimes_{\text{Free}(K(n, i-1)[p^n])} \mathbb{S}_p^\wedge = W(n+1, i-1).$$

For $i = n$, the new block is $E_n^{[n+1]} \simeq \text{Free}(\mathbb{S}_p^\wedge[x_n; p^n])$, and setting $x_n = 0$ in (9) yields exactly the quotient defining $W(n+1, n-1)$. This supplies the final quotient identity. All these comparisons are obtained from the original homotopy-pushout cubes, so they use the chosen coherent maps rather than postulated identifications in a homotopy category.

Below weight p^n the quotient algebras are unchanged. At weight p^n the quotients in (13) remove the excess summand and retain exactly the monomials in $x_0, \dots, x_i$; the non- p -power extension lemma fills the remaining weights below p^{n+1} . Therefore the strengthened polynomial condition (8) also propagates. This completes the induction. $\square$

Remark 4.8. The phrase “compatible projections” refers here to the explicit recursion $\rho_i = (\rho_{i-1}q_i, \sigma_i)$. It is not a uniqueness statement about their possible null-homotopies. In particular, at $p = 2$ some nearby groups such as $[\Sigma L(1), \mathbb{S}_p^\wedge]$ may be nonzero. The construction uses no vanishing of those groups; it uses only the obstruction groups (V_p) for extending the indicated projection maps.

5. DUALITY FOR THE MITCHELL–PRIDDY SPECTRA

We prove (V_p) by computing degreewise stable cohomotopy of the Mitchell–Priddy summands. The calculation is a use of the Segal conjecture, but one must keep track of the action of $GL_k(\mathbb{F}_p)$: an abstract nonequivariant splitting of the dual of BV_k would not suffice for the Steinberg idempotent argument.

5.1. The Steinberg idempotent. Set $V_k = (\mathbb{F}_p)^k$ and $G_k = GL_k(\mathbb{F}_p)$. Put

$$Y_k = (\Sigma_+^\infty BV_k)_p^\wedge.$$

The group G_k acts on Y_k by automorphisms of V_k . Denote by $\mathfrak{S}_k \subset G_k$ the permutation matrices and by B_k the upper triangular matrices. In $\mathbb{Z}_{(p)}[G_k]$ write

$$S_k = \sum_{\sigma \in \mathfrak{S}_k} \text{sgn}(\sigma)\sigma, \quad B_k^+ = \sum_{b \in B_k} b, \quad c_k = \prod_{r=1}^k (p^r - 1).$$

Since c_k is prime to p , we can form the Steinberg idempotents

$$e_k = c_k^{-1} S_k B_k^+, \quad \bar{e}_k = c_k^{-1} B_k^+ S_k.$$

The Steinberg relations make both expressions idempotent; see [San20, §2.1]. Let

$$\text{St}_k = \mathbb{Z}_p[G_k]e_k, \quad M(k) = e_k Y_k, \quad M(0) = \mathbb{S}_p^\wedge.$$

For a left $\mathbb{Z}_p[G_k]$ -module N , projectivity of St_k and the idempotent description give

$$e_k N \cong \text{Hom}_{G_k}(\text{St}_k, N). \quad (14)$$

The left modules $e_k N$ and $\bar{e}_k N$ are abstractly isomorphic as $\mathbb{Z}_p$ -modules (the two idempotents define isomorphic Steinberg projectives). We use this only to compute their underlying modules, not to claim that the two endomorphisms agree.

If $W \subset V_k$ has dimension r , let P_W be its stabilizer and choose a complement $V_k = W \oplus W'$. Sankar’s parabolic restriction formula identifies the restriction of St_k to P_W as the representation induced from the Levi subgroup:

$$\text{Res}_{P_W}^{G_k} \text{St}_k \cong \text{Ind}_{GL(W) \times GL(W')}^{P_W} (\text{St}_W \boxtimes \text{St}_{W'}). \quad (15)$$

This is the parabolic restriction formula of [San20, Propositions 8 and 10 in the arXiv version]. The action of P_W on the quotient V_k/W factors through $GL(W')$; in particular the $GL(W)$ factor acts trivially on a module obtained from the suspension spectrum of $B(V_k/W)$.

5.2. A natural transfer description of the dual. For a finite abelian p -group V , let $\mathcal{F}_V$ be the symmetric monoidal groupoid of finite left V -sets and equivariant bijections, with disjoint union as monoidal product. Denote its connective K -theory spectrum by $K\mathcal{F}_V$. The finite-cover transfer defines an evaluation map

$$\Delta_V : K\mathcal{F}_V \longrightarrow F(\Sigma_+^\infty BV, \mathbb{S}).$$

Proposition 5.1 (Transfer model of the Segal map). *For a finite abelian p -group V , the map obtained from Δ_V by p -completion is an equivalence*

$$(K\mathcal{F}_V)_p^\wedge \xrightarrow{\cong} F(\Sigma_+^\infty BV, \mathbb{S}_p^\wedge).$$

The equivalence is natural under automorphisms of V .

Proof. The equivariant Barratt–Priddy–Quillen theorem identifies $K\mathcal{F}_V$ with the V -fixed points of the genuine V -sphere [GM17, Theorems 5.9 and 6.2]. In this description Δ_V is the map from genuine fixed points to Borel fixed points. The Segal conjecture identifies the latter with completion at the augmentation ideal of the Burnside ring. For a p -group, the comparison between augmentation-ideal completion and p -completion is described by [Rag11, Proposition 6.4], and yields the stated p -complete map on the transfer summands. All maps entering the construction are natural for isomorphisms of finite groups, so the resulting equivalence is natural under $\text{Aut}(V)$. $\square$

Because V is abelian, every orbit in a finite V -set has the form V/H for a unique subgroup $H \leq V$. The category of finite V -sets splits into orbit types; an object consisting of m copies of V/H has automorphism group $(V/H) \wr \Sigma_m$. The nonequivariant Barratt–Priddy–Quillen theorem, applied separately to each orbit type, therefore gives

$$K\mathcal{F}_V \simeq \bigvee_{H \leq V} \Sigma_+^\infty B(V/H). \quad (16)$$

This splitting has the necessary naturality: an automorphism g of V sends the H -orbit type to the gH -orbit type by the quotient isomorphism $V/H \rightarrow V/gH$. We spell out the corresponding maps to the dual to avoid relying on an unspecified choice of splitting.

Corollary 5.2 (Automorphism-equivariant dual splitting). *For $V = V_k$ and $G = G_k$ there is a G -equivariant equivalence*

$$\bigvee_{H \leq V} (\Sigma_+^\infty B(V/H))_p^\wedge \xrightarrow{\sim} F(\Sigma_+^\infty BV, \mathbb{S}_p^\wedge). \quad (17)$$

Here $g \in G$ sends the H -summand to the gH -summand using the map induced by $V/H \cong V/gH$, while on the target it acts by precomposition with Bg^{-1} .

Proof. For each $H \leq V$, write $q_H : V \rightarrow V/H$ and introduce the graph subgroup

$$\Gamma_H = \{(v, z) \in V \times V/H : q_H(v) = z\}.$$

The summand indexed by H in (17) is adjoint to the stable transfer along the finite cover

$$B\Gamma_H \longrightarrow BV \times B(V/H),$$

followed by collapse to the sphere. These are precisely the orbit-type components of the evaluation map of Proposition 5.1; hence their wedge is an equivalence.

For $g \in G$, the map $(v, z) \mapsto (gv, \bar{g}_H z)$ identifies Γ_H with Γ_{gH} . These identifications compose with the group law in G . Naturality of finite-cover transfer makes the displayed wedge equivalence G -equivariant, with no extra choice of a representative for each orbit of subgroups. $\square$

5.3. Applying the Steinberg idempotent. For $X \in \text{Sp}_p^\wedge$, put $D(X) = F(X, \mathbb{S}_p^\wedge)$. We use the left G_k -action on $N_j = [\Sigma^j Y_k, \mathbb{S}_p^\wedge]$ given by $(g \cdot f) = f \circ \Sigma^j Bg^{-1}$.

Proposition 5.3 (Steinberg summand of the dual). *For every $k \geq 1$ and $j \in \mathbb{Z}$ there is an isomorphism of $\mathbb{Z}_p$ -modules*

$$\pi_j D(M(k)) \cong \pi_j M(k) \oplus \pi_j M(k-1).$$

Proof. The anti-involution of $\mathbb{Z}_p[G_k]$ sending g to g^{-1} interchanges the two expressions for the Steinberg idempotent. Applying D to the summand $e_k Y_k$, and then passing to degree j , gives

$$\pi_j D(M(k)) \cong \bar{e}_k N_j \cong e_k N_j.$$

By Corollary 5.2, as a G_k -module

$$N_j \cong \bigoplus_{H \leq V_k} \pi_j(\Sigma_+^\infty B(V_k/H))_p^\wedge. \quad (18)$$

There are finitely many subgroups, so this is an ordinary direct sum in degree j .

We examine each G_k -orbit in the set of subspaces. Fix an r -dimensional $W_r \subset V_k$, with stabilizer P_r . The contribution of its orbit to (18) is the induced module

$$\mathrm{Ind}_{P_r}^{G_k} X_r, \quad X_r = \pi_j(\Sigma_+^\infty B(V_k/W_r))_p^\wedge.$$

By (14), Frobenius reciprocity and (15),

$$\begin{aligned} e_k \mathrm{Ind}_{P_r}^{G_k} X_r &\cong \mathrm{Hom}_{P_r}(\mathrm{Res}_{P_r}^{G_k} \mathrm{St}_k, X_r) \\ &\cong \mathrm{Hom}_{GL(W_r) \times GL(V_k/W_r)}(\mathrm{St}_r \boxtimes \mathrm{St}_{k-r}, X_r). \end{aligned}$$

The $GL(W_r)$ factor acts trivially on X_r . The Steinberg module has no trivial quotient for $r \geq 2$: modulo p it is the nontrivial irreducible Steinberg representation, and the vanishing of its coinvariants over $\mathbb{Z}_p$ follows by Nakayama. For $r = 1$ the Steinberg module of $GL_1(\mathbb{F}_p)$ is the trivial rank-one module. These observations leave exactly three boundary cases to consider:

- (i) If $r = 0$, then $W_r = 0$, $X_r = \pi_j Y_k$, and the contribution is $\pi_j M(k)$.
- (ii) If $1 \leq r \leq k-1$, all terms with $r > 1$ vanish. For $r = 1$, the surviving $GL_{k-1}(\mathbb{F}_p)$ -module is the Steinberg summand of $\pi_j \Sigma_+^\infty B V_{k-1}$; its contribution is $\pi_j M(k-1)$.
- (iii) If $r = k$, then $X_r = \pi_j \mathbb{S}_p^\wedge$ has trivial G_k -action, so the contribution is zero for $k \geq 2$. At $k = 1$ it is $\pi_j \mathbb{S}_p^\wedge = \pi_j M(0)$.

Adding the surviving orbit contributions proves the formula for all $k \geq 1$. $\square$

Theorem 5.4 (Mitchell–Priddy duality). *For every prime p , integer $k \geq 0$ and $j \in \mathbb{Z}$, there is a (not necessarily canonical) isomorphism of finitely generated $\mathbb{Z}_p$ -modules*

$$[\Sigma^j L(k), \mathbb{S}_p^\wedge] \cong \pi_j L(k) \oplus \pi_j L(k-1), \quad L(-1) = 0. \quad (19)$$

Proof. Mitchell and Priddy identify the Steinberg summand as

$$M(k) \simeq L(k) \vee L(k-1) \quad (20)$$

[AL10, Propositions 2.9 and 2.10]; the normalization is the one used in Theorem 2.5. Fix j , and set

$$a_k = \pi_j D(L(k)), \quad b_k = \pi_j L(k),$$

with $a_{-1} = b_{-1} = 0$. The spectra $L(k)$ are retracts of p -complete suspension spectra of classifying spaces of finite p -groups, so the groups in each fixed degree are finitely generated over $\mathbb{Z}_p$. Dualizing (20) and using Proposition 5.3, we get

$$a_k \oplus a_{k-1} \cong (b_k \oplus b_{k-1}) \oplus (b_{k-1} \oplus b_{k-2}).$$

For $k = 0$, the assertion is $D(\mathbb{S}_p^\wedge) \simeq \mathbb{S}_p^\wedge$. Assume by induction that $a_{k-1} \cong b_{k-1} \oplus b_{k-2}$. Cancellation for finitely generated modules over the principal ideal domain $\mathbb{Z}_p$ gives $a_k \cong b_k \oplus b_{k-1}$. Since $a_k = [\Sigma^j L(k), \mathbb{S}_p^\wedge]$, this proves (19). $\square$

Corollary 5.5 (The vanishing required in the construction). *Condition (V_p) holds for every prime p :*

$$[\Sigma^{k-2} L(k), \mathbb{S}_p^\wedge] = 0 \quad (k \geq 2).$$

More generally $[\Sigma^j L(k), \mathbb{S}_p^\wedge] = 0$ for $j < 0$ and $k \geq 0$.

Proof. By Theorem 5.4, the first group is the direct sum

$$\pi_{k-2}L(k) \oplus \pi_{k-2}L(k-1).$$

Both summands vanish by Lemma 2.6, since $k-2 < c(k)$ and $k-2 < c(k-1)$. For negative j , connectivity of $L(k)$ and $L(k-1)$ gives the last assertion; when $k=0$ it is connectivity of the sphere. $\square$

6. ASSEMBLY AND SPHERICAL LIFTS

We first take the limit of the weightwise construction. The resulting algebra has a p -complete underlying ungraded ring, but keeping its uncompleted graded direct sum until the final step is useful: it shows exactly which elements are needed to identify the δ_p -structure.

Theorem 6.1 (The graded lift). *There exists a connected graded E_∞ -algebra*

$$\mathcal{R} \in \text{CAlg}(\text{Gr}(\text{Sp}_p^\wedge))$$

with $\mathcal{R}_0 \simeq \mathbb{S}_p^\wedge$ and every $\mathcal{R}_w$ finite free, together with classes $x_i \in \pi_0 \mathcal{R}_{p^i}$ satisfying

$$H\mathbb{F}_p \otimes \mathcal{R} \simeq H(\mathbb{F}_p[x_0, x_1, \dots]), \quad |x_i| = p^i, \quad \delta_p(x_i) = x_{i+1}.$$

In particular,

$$\pi_0\left(\bigoplus_{w \geq 0} \mathcal{R}_w\right) \cong \mathbb{Z}_p[x_0, x_1, \dots].$$

Proof. By Corollary 5.5, condition (V_p) holds, so we may apply Theorem 4.7 at every stage starting from the free weight-one algebra. Let $R^{(n)}$ be the algebra at the end of stage n , and write

$$T_n = \tau_{<p^{n+1}} R^{(n)}.$$

The attachment at stage $n+1$ takes place first in weight p^{n+1} ; thus the chosen stage maps induce equivalences

$$\tau_{<p^{n+1}} T_{n+1} \simeq T_n.$$

For any fixed m , choose n large enough that $m < p^{n+1}$. The truncation $\tau_{\leq m} T_n$ is then independent of further stages, in the sense of the specified compatible equivalences. These truncated algebras form an inverse system, and we set

$$\mathcal{R} = \varprojlim_m \tau_{\leq m} T_{n(m)}.$$

Weight truncation commutes with this limit, so in a fixed weight the limiting component is exactly the component constructed at a sufficiently late finite stage. It is therefore finite free, and its reduction modulo p has the asserted monomial basis. Passing through each weight gives the displayed equivalence of graded Eilenberg–Mac Lane algebras. The multiplicative map from the polynomial ring to the graded π_0 is an isomorphism modulo p in every weight. Since both sides are finite free over $\mathbb{Z}_p$ in each weight, Nakayama shows that it is an isomorphism integrally.

For fixed i , the identity $\delta_p(x_i) = x_{i+1}$ was imposed when weight p^{i+1} was attached. Later stages leave this weight unchanged, and the Frobenius construction is natural. The identity therefore persists in the limit. $\square$

Theorem 6.2 (The local Moore lift). *Let $A_p = (\mathbb{Z}_p\{x\})_p^\wedge$. There is a p -complete Moore E_∞ -ring R_p with*

$$\pi_0 R_p \cong A_p$$

as δ_p -rings. As a p -complete spectrum, R_p can be chosen in the form

$$R_p \simeq \left(\bigvee_{\alpha \in \mathcal{I}} \mathbb{S}_p^\wedge \right)_p^\wedge,$$

where $\mathcal{I}$ is the countable set of monomials in the variables x_i .

Proof. Forget the grading on the algebra of Theorem 6.1 and set

$$U = \bigoplus_{w \geq 0} \mathcal{R}_w, \quad R_p = U_p^\wedge.$$

Each weight- w component has a sphere summand for every monomial of weight w , so the underlying spectrum of U is a wedge indexed by all monomials. The identification of its degree-zero homotopy ring in Theorem 6.1 yields

$$\pi_0 R_p \cong (\mathbb{Z}_p[x_0, x_1, \dots])_p^\wedge.$$

Moreover $H\mathbb{F}_p \otimes R_p$ is discrete. By [CNY26, Lemma 2.40], $R_p \otimes H\mathbb{Z}$ is the discrete Eilenberg–Mac Lane spectrum of this completed ring, proving that R_p is a Moore spectrum.

By Proposition 3.1, the δ_p -operation on $\pi_0 R_p$ is the continuous extension of the homogeneous operation on $\pi_0 U$. Its values on generators are $\delta_p(x_i) = x_{i+1}$ by Theorem 6.1. The coefficient ring $\mathbb{Z}_p$ has its usual p -derivation, for which the Frobenius lift is the identity. Consequently the comparison $\mathbb{Z}_p\{x\} \rightarrow \pi_0 U$ is an isomorphism of δ_p -rings already before completion; the same holds after completion. Alternatively one may apply the criterion of Lemma 3.2, with all coefficients $c_i = 1$. $\square$

6.1. Arithmetic gluing. An integral lift does not require a prescribed δ_ℓ -structure for $\ell \neq p$. We use this freedom to make the other local factors particularly simple and then apply an arithmetic pullback in commutative ring spectra.

Theorem 6.3 (An integral Moore lift). *For each fixed prime p there is a Moore E_∞ -ring $R_\mathbb{Z}$ and a ring isomorphism*

$$\theta : \mathbb{Z}[x_0, x_1, \dots] \xrightarrow{\cong} \pi_0 R_\mathbb{Z}$$

such that $(R_\mathbb{Z})_p^\wedge \simeq R_p$ and the induced δ_p -structure on the p -completion is the free one. No condition on the δ_ℓ -operations for $\ell \neq p$ is asserted.

Proof. At the selected prime p take R_p from Theorem 6.2. For each other prime ℓ , take the ℓ -complete spherical monoid ring on countably many generators,

$$R_\ell = \mathbb{S}[\mathbb{N}^{(\mathbb{N})}] \wedge_\ell.$$

It is a Moore E_∞ -ring and $\pi_0 R_\ell \cong (\mathbb{Z}_\ell[x_0, x_1, \dots]) \wedge_\ell$; compare [CNY26, Construction 2.3 and Lemma 2.40]. Each R_ℓ is connective and has, as underlying spectrum, an ℓ -completed wedge of spheres indexed by monomials.

Put

$$\Pi = \prod_\ell R_\ell, \quad \Pi_\mathbb{Q} = \Pi \otimes H\mathbb{Q}, \quad F = H\mathbb{Q}[x_0, x_1, \dots].$$

For every $r > 0$, the stable homotopy group $\pi_r \mathbb{S}$ is finite. Only finitely many primes can contribute in that degree, and the exponents of their torsion are bounded in terms of r . This remains true for any completed wedge of sphere summands, because completion cannot

increase the exponent of a bounded torsion group. Hence $\pi_r \Pi$ is bounded torsion for $r > 0$, and

$$\pi_r \Pi_{\mathbb{Q}} = 0 \quad (r \neq 0).$$

Thus $\Pi_{\mathbb{Q}}$ is a rational discrete commutative ring spectrum. Sending x_i to the tuple of the corresponding local classes defines a map of commutative $H\mathbb{Q}$ -algebras

$$a : F \longrightarrow \Pi_{\mathbb{Q}}.$$

Now form the homotopy pullback in $\mathrm{CAlg}(\mathrm{Sp})$:

$$R_{\mathbb{Z}} = F \times_{\Pi_{\mathbb{Q}}} \Pi. \quad (21)$$

We check its homotopy in degree zero rather than appealing only to a general fracture-square slogan. Let $A_{\ell} = \pi_0 R_{\ell}$. The homotopy pullback yields the exact sequence

$$0 \longrightarrow \pi_0 R_{\mathbb{Z}} \longrightarrow \mathbb{Q}[x_0, x_1, \dots] \oplus \prod_{\ell} A_{\ell} \xrightarrow{\alpha} \left(\prod_{\ell} A_{\ell} \right) \otimes \mathbb{Q} \longrightarrow \pi_{-1} R_{\mathbb{Z}} \longrightarrow 0, \quad (22)$$

where $\alpha(q, (s_{\ell})) = (a_{\ell}(q) - s_{\ell})_{\ell}$. It is surjective. Indeed, an arbitrary element of the target is represented by $(r_{\ell})_{\ell}/N$ for a positive integer N . For each $\ell \mid N$ choose a polynomial P_{ℓ} approximating r_{ℓ} modulo $\ell^{v_{\ell}(N)}$. Such a polynomial exists because A_{ℓ} is the ℓ -adic completion of a polynomial ring, so its elements are approximated modulo any fixed ℓ -power by finite polynomials. Apply the Chinese remainder theorem coefficient by coefficient to the finite family of primes dividing N and choose an integral polynomial Q with

$$Q \equiv r_{\ell} \pmod{\ell^{v_{\ell}(N)}} \quad (\ell \mid N).$$

Then $(Q - r_{\ell})/N$ belongs to A_{ℓ} for every ℓ , including those not dividing N , and

$$\alpha\left(Q/N, ((Q - r_{\ell})/N)_{\ell}\right) = (r_{\ell})_{\ell}/N.$$

Therefore $\pi_{-1} R_{\mathbb{Z}} = 0$.

The kernel of α consists of rational polynomials integral in every ℓ -adic coefficient ring. Writing $q = Q/N$, coefficientwise integrality for all $\ell \mid N$ says that each coefficient of Q is divisible by N . Thus

$$\ker \alpha = \mathbb{Z}[x_0, x_1, \dots], \quad \pi_0 R_{\mathbb{Z}} \cong \mathbb{Z}[x_0, x_1, \dots].$$

Since F and $\Pi_{\mathbb{Q}}$ have no positive homotopy, the same long exact sequence shows that $\pi_r R_{\mathbb{Z}} \cong \pi_r \Pi$ for $r > 0$.

We next verify the Moore condition, for which the preceding computation of π_0 alone is insufficient. Smashing spectra with $H\mathbb{Z}$ is exact, so it preserves the underlying homotopy pullback square (21). In addition, the natural map

$$H\mathbb{Z} \otimes \left(\prod_{\ell} R_{\ell} \right) \longrightarrow \prod_{\ell} (H\mathbb{Z} \otimes R_{\ell}) \quad (23)$$

is an equivalence. To justify the latter, fix a degree d . A finite cell approximation to $H\mathbb{Z}$ can be chosen with cofiber $(d+1)$ -connective. Smashing with any connective R_{ℓ} preserves this connectivity uniformly in ℓ . A finite spectrum is dualizable, hence its smash product commutes with arbitrary products. Comparing homotopy in degrees $\leq d$ on the two sides of (23) gives an isomorphism; now let d vary. Each $H\mathbb{Z} \otimes R_{\ell}$ is discrete, as are $H\mathbb{Z} \otimes F$ and $H\mathbb{Z} \otimes \Pi_{\mathbb{Q}}$. The pullback of these Eilenberg–Mac Lane spectra is discrete because the map α in (22) is surjective. Its degree-zero group is the polynomial ring computed above. Hence

$$H\mathbb{Z} \otimes R_{\mathbb{Z}} \simeq H\mathbb{Z}[x_0, x_1, \dots],$$

proving that $R_{\mathbb{Z}}$ is a Moore E_{∞} -ring.

Finally, derived p -completion of (21) recovers the p -component R_p . Indeed, the two rational terms become zero after p -completion, while the factors R_{ℓ} for $\ell \neq p$ have multiplication by p invertible and hence have trivial p -completion. The projection $R_{\mathbb{Z}} \rightarrow R_p$ carries x_i to x_i . Naturality of the operation of [CNY26, Proposition 2.41] and Theorem 6.2 show that its completed p -derivation sends x_i to x_{i+1} , as required. $\square$

Proof of Theorem 1.1. The local statement, including the compatible graded refinement, is Theorems 6.1 and 6.2. The integral statement is Theorem 6.3. $\square$

Remark 6.4. The construction concerns existence and the prescribed Tate–Frobenius operation. It does not assert that the spherical lift is unique, or that its mapping spaces satisfy a universal property resembling that of the free δ_p -ring. Such a statement would require a separate analysis of maps between the constructed E_{∞} -rings.

REFERENCES

- [AD01] G. Arone and W. Dwyer, *Partition complexes, Tits buildings and symmetric products*, Proc. London Math. Soc. (3) **82** (2001), no. 1, 229–256.
- [AL10] G. Arone and K. Lesh, *Augmented Γ -spaces, the stable rank filtration, and a bu analogue of the Whitehead conjecture*, Fund. Math. **207** (2010), no. 1, 29–70.
- [CNY26] S. Carmeli, T. Nikolaus, and A. Yuan, *Maps between spherical group rings*, Invent. Math. (2026), doi:10.1007/s00222-026-01446-0.
- [GM17] B. Guillou and J. P. May, *Equivariant iterated loop space theory and permutative G -categories*, Algebr. Geom. Topol. **17** (2017), no. 6, 3259–3339.
- [Kuh15] N. J. Kuhn, *The Whitehead conjecture, the tower of S^1 conjecture, and Hecke algebras of type A*, J. Topol. **8** (2015), no. 1, 118–146.
- [Lur17] J. Lurie, *Higher Algebra*, 2017, available from the author’s webpage.
- [MP83] S. A. Mitchell and S. B. Priddy, *Stable splittings derived from the Steinberg module*, Topology **22** (1983), no. 3, 285–298.
- [Nak58] M. Nakaoka, *Cohomology mod p of symmetric products of spheres*, J. Inst. Polytech. Osaka City Univ. Ser. A **9** (1958), 1–18.
- [NS18] T. Nikolaus and P. Scholze, *On topological cyclic homology*, Acta Math. **221** (2018), no. 2, 203–409.
- [Rag11] K. Ragnarsson, *Completion of G -spectra and stable maps between classifying spaces*, Adv. Math. **227** (2011), no. 4, 1539–1561.
- [San20] K. Sankar, *Equivariant Steinberg summands*, Homology Homotopy Appl. **22** (2020), no. 2, 203–220; arXiv:1903.08246 (earlier numbering).